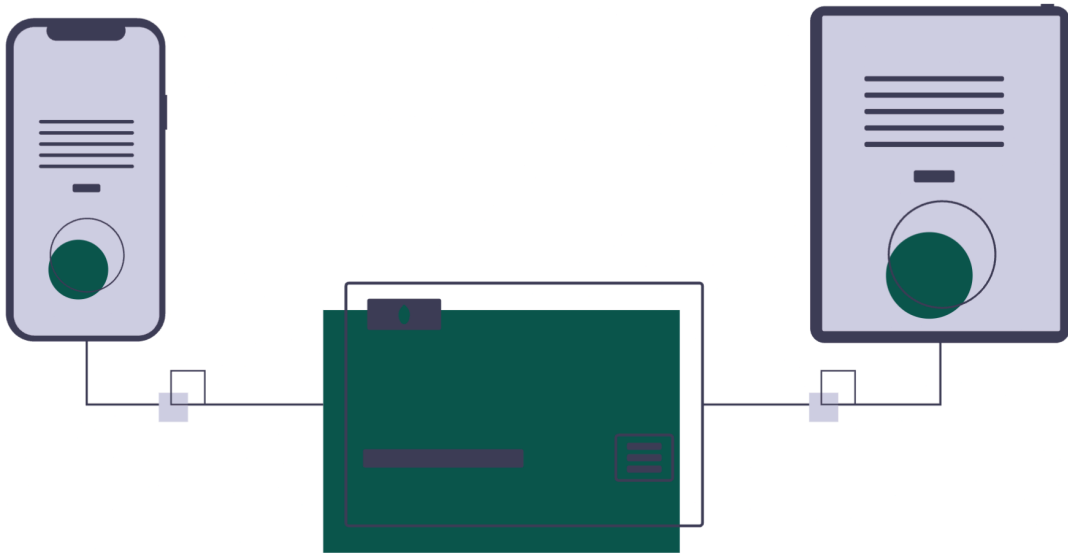


Lab Final Assignment of CSE405

Computer Networks



Submitted By:

Aklhak Hossain

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<https://akhlak.com>

Submitted To:

Md. Khalid Mahbub Khan

Senior Lecturer

Department of CSE

East West University

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Objective:

The goal of this task is to build a fully fault-tolerant network for a university spread across five campuses. The design incorporates a primary linear path, loop connections to provide alternate routes, and cross-links between routers to guarantee that the network continues operating even when a single connection fails. All host IP addresses are assigned manually, with no dynamic routing protocols involved.

Network Architecture

Based on the provided simulation layout, the topology is structured as follows:

Five Campus Routers represent each of the five university locations. Each router connects to a local switch, which in turn supports three PCs, giving a total of 15 end-user hosts across the network.

Resilient Connections: The routers are interlinked in a loop arrangement, establishing an alternate forwarding path that activates automatically if any primary link goes down. Additional cross-connections between non-adjacent routers create a star-mesh layout, resulting in a highly robust and redundant topology that can withstand multiple simultaneous link failures.

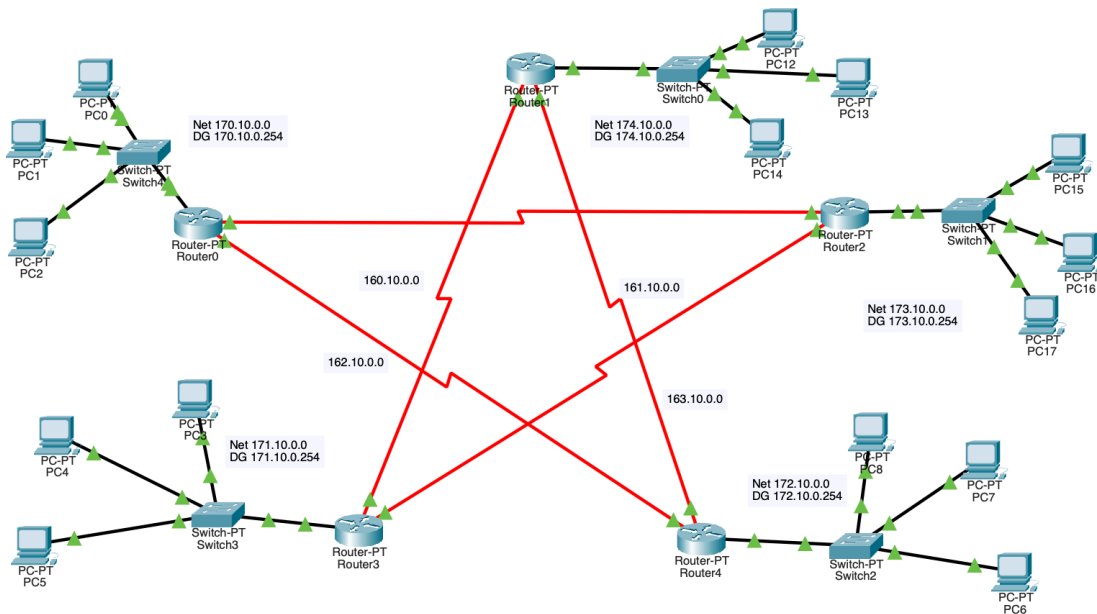


Figure: Network Topology

IP Addressing Scheme:

In line with the task requirements, Class B IP addresses are used exclusively throughout the entire topology. Network addresses and their corresponding default gateways have been assigned in a logical and organized manner to clearly separate each campus segment.

Router	Interface	IP Address	Subnet Mask
Router 0	fa0/0	170.10.0.254	255.255.0.0
	se2/0	162.10.0.1	255.255.0.0
	se3/0	164.10.0.2	255.255.0.0
Router 1	fa0/0	174.10.0.254	255.255.0.0
	se2/0	161.10.0.1	255.255.0.0
	se3/0	160.10.0.2	255.255.0.0
Router 2	fa0/0	173.10.0.254	255.255.0.0
	se2/0	163.10.0.1	255.255.0.0
	se3/0	164.10.0.2	255.255.0.0
Router 3	fa0/0	171.10.0.254	255.255.0.0
	se2/0	160.10.0.1	255.255.0.0
	se3/0	163.10.0.2	255.255.0.0
Router 4	fe0/0	172.10.0.254	255.255.0.0
	se2/0	162.10.0.1	255.255.0.0
	se3/0	161.10.0.2	255.255.0.0

Table : Network & Host IP Addresses

Conclusion

The completed network design successfully meets all requirements for a large-scale, resilient university infrastructure. The structured use of Class B addressing gives each campus network well-defined operational boundaries, while the physically redundant inter-router connections ensure that any single link failure can be transparently bypassed through an alternative path, keeping all campuses continuously reachable.